IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- ❑ Executive Summary
- ❑ Introduction
- ❑ Methodology
- ❑ Results
- ❑ Conclusion
- ❑ Appendix

Executive Summary

❑ Summary of Methodologies:

- SpaceX Data Collection using SpaceX API
- SpaceX Data Collection with Web Scraping
- SpaceX Data Wrangling
- SpaceX Exploratory Data Analysis using SQL
- SpaceX EDA with Visualization using Python, Pandas and Matplotlib
- SpaceX Interactive Visual Analytics with Folium and Plotly Dash
- SpaceX Machine Learning Landing Prediction

❑ Summary of all Results

- EDA Results
- Interactive Visual Analytics and Dashboards

Introduction

❑ Project Background And Context

❖ SpaceX advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars; other providers cost upward of 165 million dollars each, much of the savings is because SpaceX can reuse the first stage. Therefore if we can determine if the first stage will land, we can determine the cost of a launch. This information can be used if an alternate company wants to bid against SpaceX for a rocket launch.

❑ Problems You Want To Find Answers

❖ In this capstone, we will predict if the Falcon 9 first stage will land successfully using Data from Falcon9 rocket launches advertised on its website.

Section 1

Methodology

Methodology

Executive Summary

- ❑ **Data Collection Methodology**
  - Describe how data was collected
- ❑ **Perform Data Wrangling**
  - Describe how data was processed
- ❑ **Perform Exploratory Data Analysis (EDA) using Visualization and SQL**
- ❑ **Perform Interactive Visual Analytics using Folium and Plotly Dash**
- ❑ **Perform Predictive Analysis using Classification Models**
  - How to Build, Tune, Evaluate Classification Models

Data Collection

Collecting SpaceX Falcon9 Data

Data was first collected using SpaceX API (a RESTful API) by making a get request to the SpaceX API. This was done by first defining a series helper functions that would help in the use of the API to extract information using identification numbers in the launch data and then requesting rocket launch data from the SpaceX API URL.

Data Collection – SpaceX API

Data was first collected using SpaceX API (a RESTful API) by making a get request to the SpaceX API. This was done by first defining a series helper functions that would help in the use of the API to extract information using identification numbers in the launch data and then requesting rocket launch data from the SpaceX API URL. Finally, to make the requested JSON results more consistent, the SpaceX launch data was requested and parsed using the GET request and then decoded the response content as a Json result which was then converted into a Pandas data frame.

Task 1: Request and parse the SpaceX launch data using the GET request

To make the requested JSON results more consistent, we will use the following static response object for this project:

```
static_json_url='https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/API_call_spacex_api.json'
```

We should see that the request was successfull with the 200 status response code

```
response=requests.get(static_json_url)
```

```
response.status_code
```

```
200
```

Now we decode the response content as a Json using `.json()` and turn it into a Pandas dataframe using `.json_normalize()`

```
# Use json_normalize meethod to convert the json result into a dataframe
respjson = response.json()
data = pd.json_normalize(respjson)
```

GITHUB URL: https://github.com/swatigangotri31-rgb/SpaceX-Falcon9-1st-Stage-Success-Landing-Predictions.git

Data Collection - Scraping

To perform web scraping, collected Falcon 9 historical launch records from a Wikipedia page titled List of Falcon 9 and Falcon Heavy launches of the launch records are stored in a HTML. Using Beautiful Soup and request Libraries, extract the Falcon 9 launch HTML table records from the Wikipedia page, Parsed the table, and converted it into a Pandas data frame

GITHUB URL: https://github.com/swatigangotri31-rgb/SpaceX-Falcon9-1st-Stage-Success-Landing-Predictions.git

TASK 1: Request the Falcon9 Launch Wiki page from its URL

First, let's perform an HTTP GET method to request the Falcon9 Launch HTML page, as an HTTP response.

```
# use requests.get() method with the provided static_url and headers
# assign the response to a object
response = session.get(static_url, headers=headers)

print("Status:", response.status_code)
print(response.text[:300])  # debug
```

```
Status: 200
<!DOCTYPE html>
<html class="client-nojs vector-feature-language-in-header-enabled vector-feature-language-in-main-menu-disabled vector-feature-language-in-main-page-header-disabled
vector-feature-page-tools-pinned-disabled vector-feature-toc-pinned-clientpref-1 vector-feature-main-menu-pinned-disab
```

Create a `BeautifulSoup` object from the HTML `response`

```
# Use BeautifulSoup() to create a BeautifulSoup object from a response text content
soup = BeautifulSoup(response.content, 'html.parser')
print(response.status_code)
print(response.text[:300])
```

```
200
<!DOCTYPE html>
<html class="client-nojs vector-feature-language-in-header-enabled vector-feature-language-in-main-menu-disabled vector-feature-language-in-main-page-header-disabled
vector-feature-page-tools-pinned-disabled vector-feature-toc-pinned-clientpref-1 vector-feature-main-menu-pinned-disab
```

Print the page title to verify if the `BeautifulSoup` object was created properly

```
# Use soup.title attribute
print(soup.title)
```

```
<title>List of Falcon 9 and Falcon Heavy launches - Wikipedia</title>
```

Data Wrangling

After obtaining and creating a Pandas DF from the collected data, data was filtered using the BOOSTERVERSION column to only keep the Falcon 9 launches, then dealt with the missing data values in the LANDINGPAD and PAYLOADMASS columns. For the PayloadMass ,missing data values were replaced using mean value of column. Also performed some Exploratory Data Analysis (EDA) to find some patterns in the data and determine what would be the label for training supervised models.

GITHUB
URL: https://github.com/swatigangotri31-rgb/SpaceX-Falcon9-1st-Stage-Success-Landing-Predictions.git

TASK 4: Create a landing outcome label from Outcome column

Using the `Outcome`, create a list where the element is zero if the corresponding row in `Outcome` is in the set `bad_outcome`; otherwise, it's one. Then assign it to the variable `landing_class`:

```
# landing_class = 0 if bad_outcome
df['Class'] = df['Outcome'].apply(lambda x: 0 if x in bad_outcomes else 1)
df['Class'].value_counts()
# landing_class = 1 otherwise
```

```
Class
1    60
0    30
Name: count, dtype: int64
```

This variable will represent the classification variable that represents the outcome of each launch. If the value is zero, the first stage did not land successfully; one means the first stage landed Successfully

```
landing_class=df['Class']
df[['Class']].head(8)
```

	Class
0	0
1	0
2	0
3	0
4	0
5	0
6	1
7	1

EDA with Data Visualization

- The relationship between Flight Number and Launch Site -> scatter plot
- The relationship between Payload and Launch Site -> scatter plot
- The relationship between success rate of each orbit type -> bar plot
- The relationship between Flight Number and Orbit type -> scatter plot
- The relationship between Payload and Orbit type -> scatter plot
- The launch success yearly trend -> line chart
- The scatter plot is the best to describe the relation between two categorical data
- The bar plot is the best to compare several categorical data
- The line plot is the best to show the time series data

- GITHUB URL: https://github.com/swatigangotri31-rgb/SpaceX-Falcon9-1st-Stage-Success-Landing-Predictions.git

EDA with SQL

- Display the names of the unique launch sites in the space mission
- Display 5 records where launch sites begin with the string 'CCA'
- Display the total payload mass carried by boosters launched by NASA (CRS)
- Display average payload mass carried by booster version F9 v1.1
- List the date when the first successful landing outcome in ground pad was achieved.
- List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000
- List the total number of successful and failure mission outcomes
- List the names of the booster_versions which have carried the maximum payload mass.
- List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.
- Rank the count of successful landing_outcomes between the date 04-06-2010 and 20-03-2017 in descending order.

- GITHUB URL: https://github.com/swatigangotri31-rgb/SpaceX-Falcon9-1st-Stage-Success-Landing-Predictions.git

Build an Interactive Map with Folium

- ❑ Mark all launch sites on a map
- ❑ Mark the success/failed launches for each site on the map
- ❑ Calculate the distances between a launch site to its proximities
- ❑ Whether it is close to the coast
- ❑ Whether it is close to the railway
- ❑ Whether it is close to the highway
- ❑ Whether it is close to the city

- ❑ GITHUB URL: https://github.com/swatigangotri31-rgb/SpaceX-Falcon9-1st-Stage-Success-Landing-Predictions.git

Build a Dashboard with Plotly Dash

- ❑ A launch site drop-down input component
- ❑ A success-pie-chart based on the selected site dropdown
- ❑ Arrange slicer to select payload
- ❑ A success-payload-scatter-chart scatter plot based on the selected site dropdown

- ❑ GITHUB URL: https://github.com/swatigangotri31-rgb/SpaceX-Falcon9-1st-Stage-Success-Landing-Predictions.git

Predictive Analysis (Classification)

- After loading the data as a Pandas DataFrame, I performed Exploratory Data Analysis and determine Training Labels by:

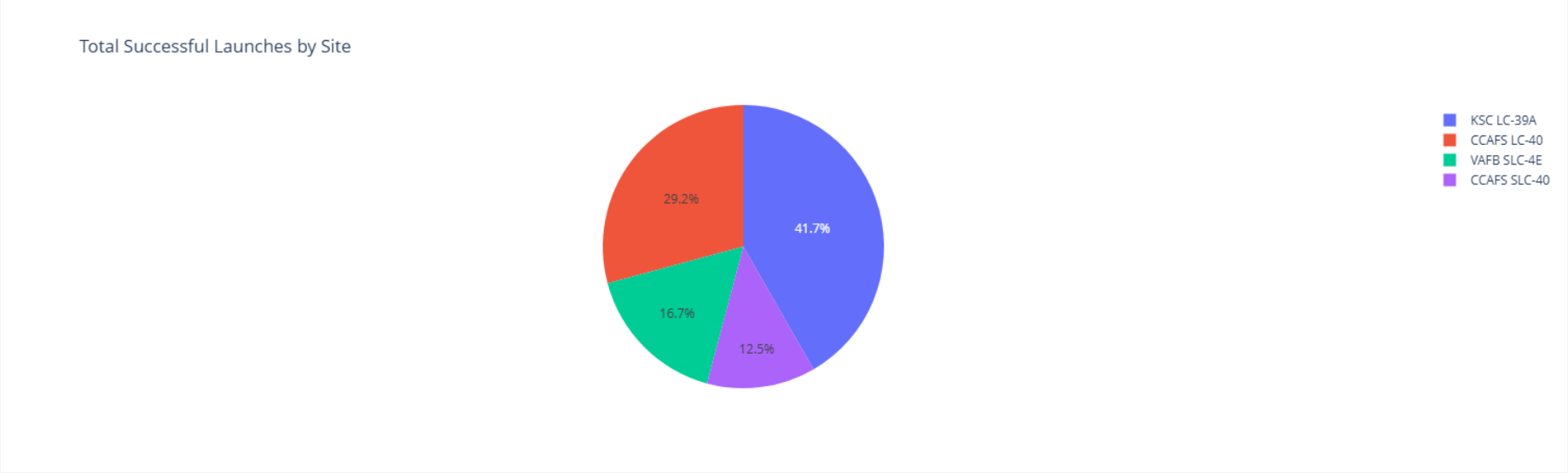
- Creating a NumPy array from the column Class in data, by applying the method to_numpy() then assigned it to the variable Y as the outcome variable.
- Then standardized the feature dataset (x) by transforming it using preprocessing.StandardScaler() function from Sklearn.
- After which the data was split into training and testing sets using the function train_test_split from sklearn.model_selection with the test_size parameter set to 0.2 and random _state to 2.

Results

EDA

KSC LC-39A and VAFB SLC 4E has a success rate of 77% + VAFB SLC 4E has no payload above 10000 kg. In the LEO orbit the Success appears related to the number of flights. With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS. The success rate since 2013 kept increasing till 2020.

Interactive Analysis



Results

Predictive Analysis

According to the decision tree classifier model, the predictive model tells us there will be 4 true positive, 7 true negative, 5 false positive and 2 negative . The accuracy of the mode is around 89% with the best parameters.

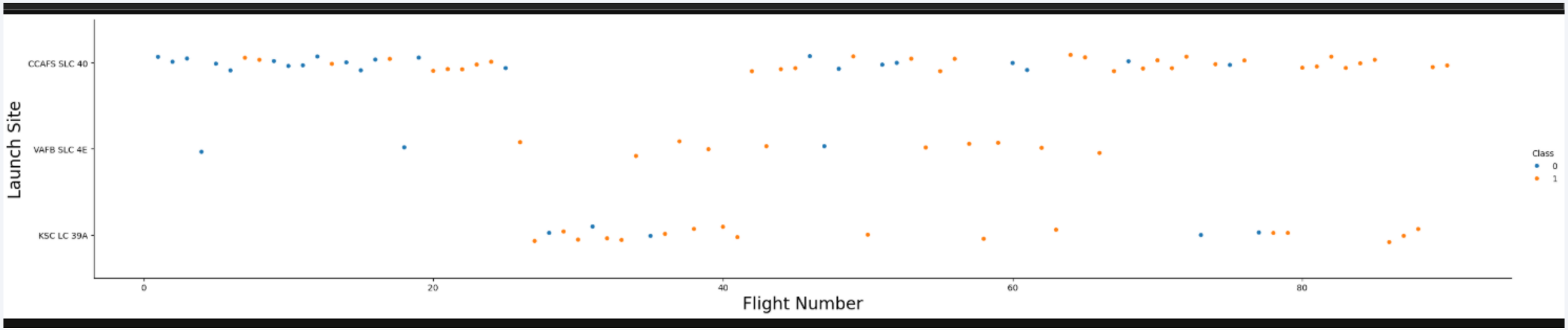
GITHUB URL: https://github.com/swatigangotri31-rgb/SpaceX-Falcon9-1st-Stage-Success-Landing-Predictions.git

Section 2

Insights drawn from EDA

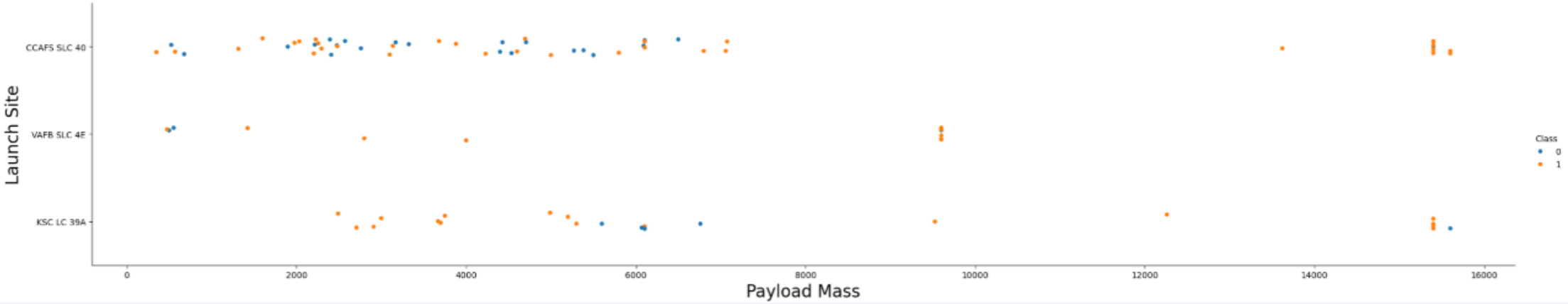
Flight Number vs. Launch Site

- Visualizing the relationship between Flight Number and Launch Site by Scatter Plot
- KSC SLC 39A has the highest successful rate, and CCAFS SLC 40 has the lowest



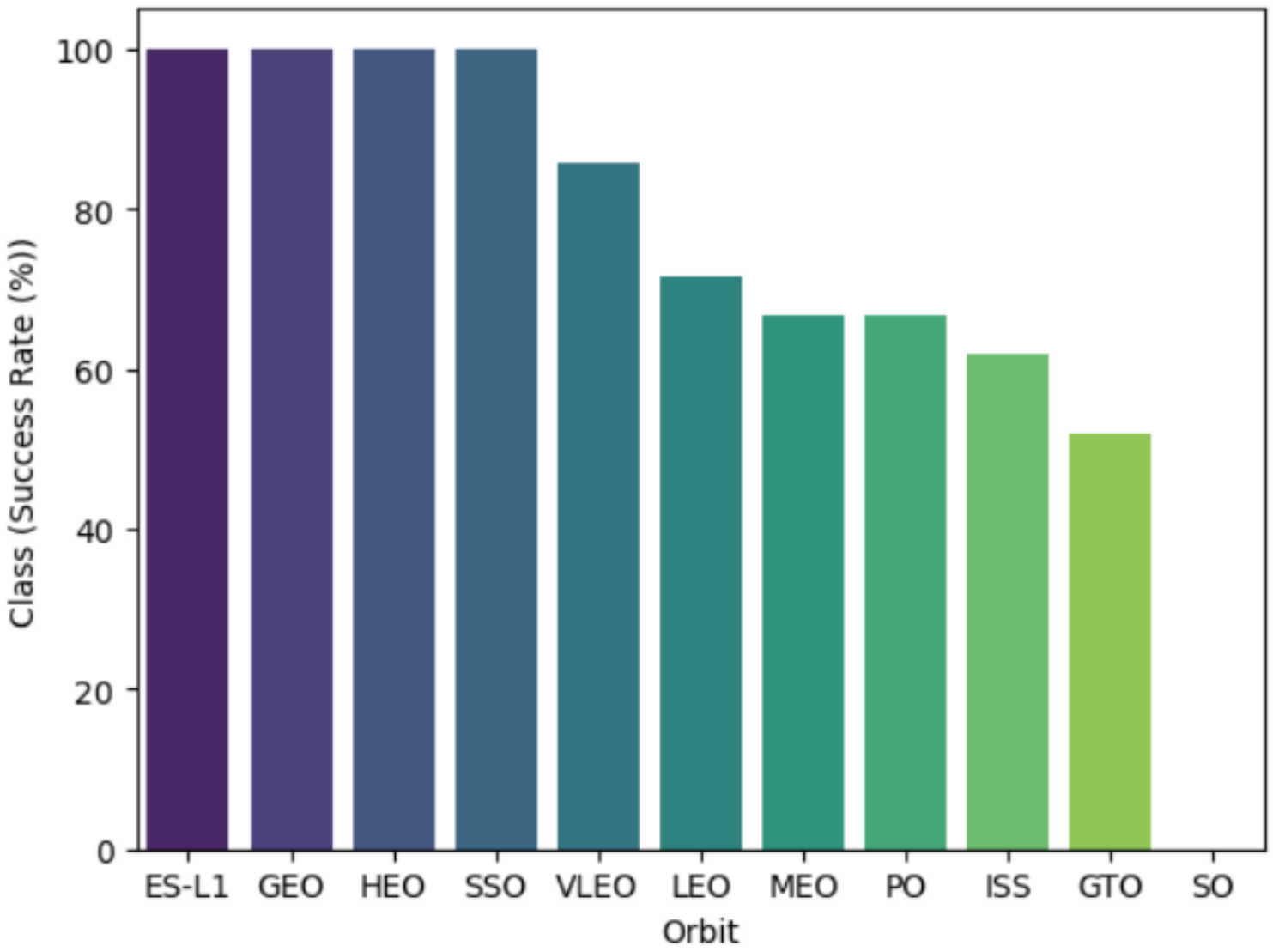
Payload vs. Launch Site

- ❑ Visualizing the relationship between Payload Mass and Launch Site by Scatter Plot
- ❑ VAFB SLC 4E has no Payload above 10000kg



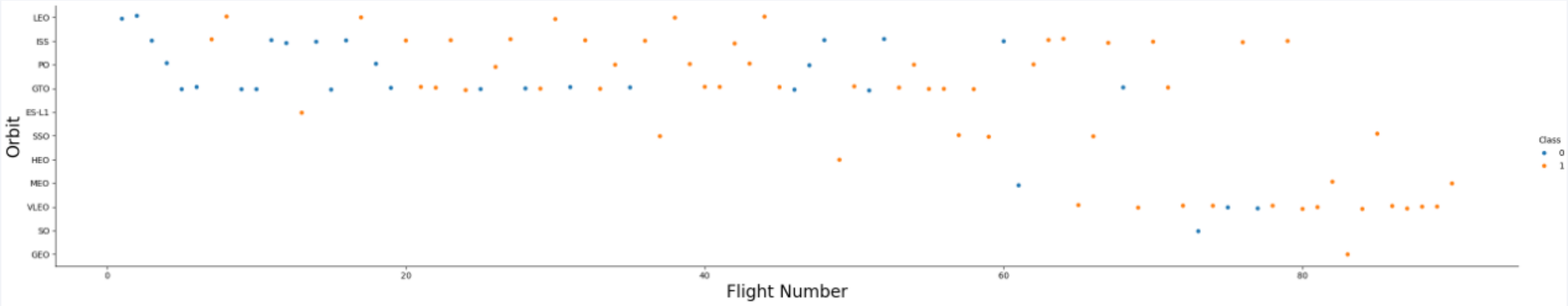
Success Rate vs. Orbit Type

- Visualizing the relationship between Success rate of each Orbit Type by Bar Plot
- ES-L1, GEO, HEO, and SSO has the highest success rate



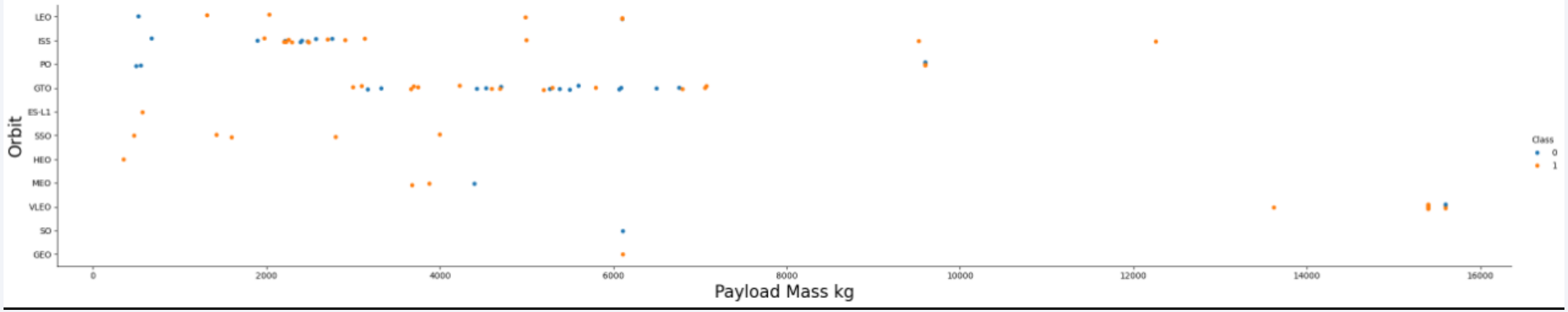
Flight Number vs. Orbit Type

- Visualizing the relationship between Flight Number and Orbit Type by Scatter Plot
- In the LEO orbit, success seems to be related to the number of flights. Conversely, in the GTO orbit, there appears to be no relationship between flight number and success.



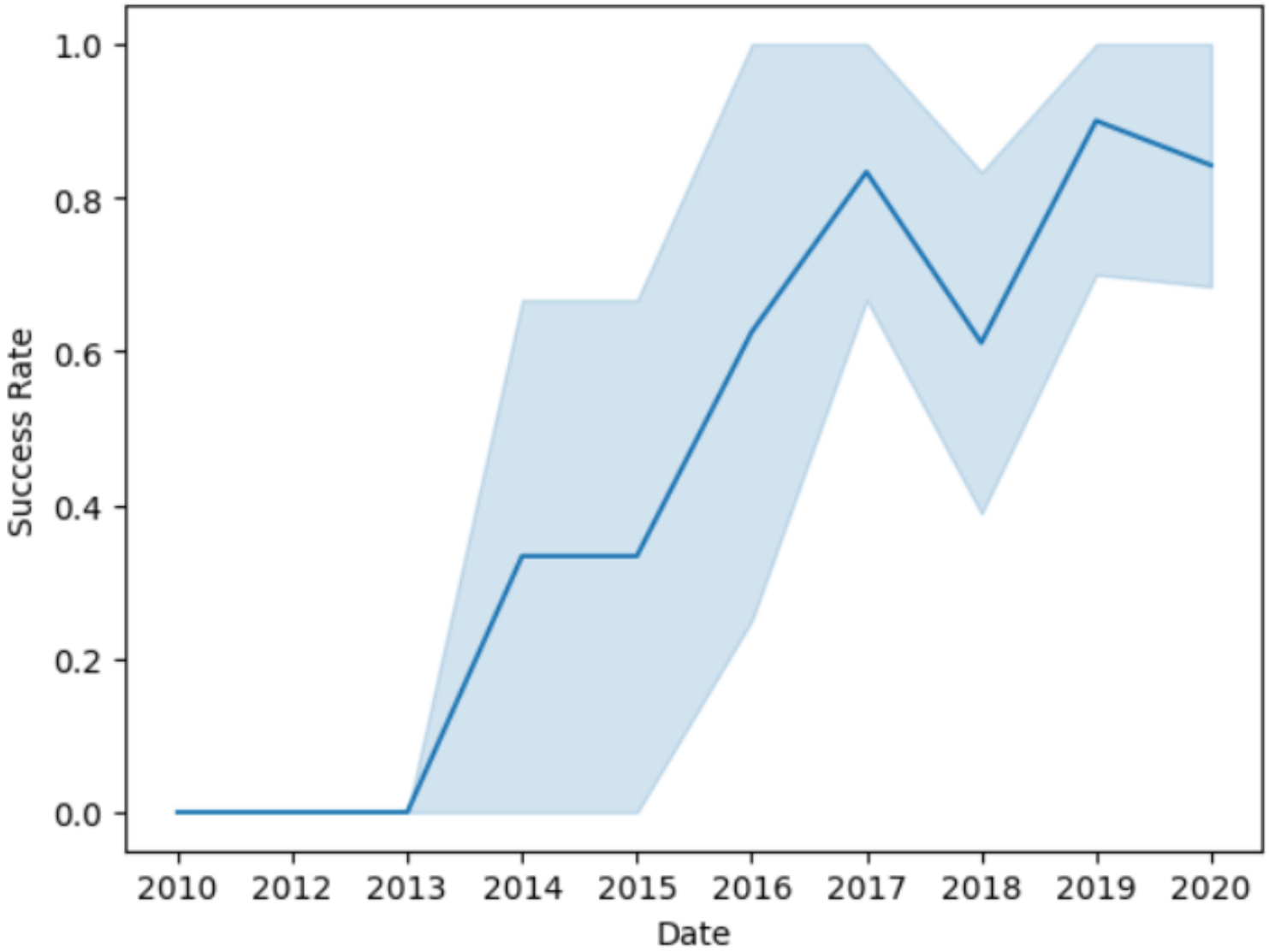
Payload vs. Orbit Type

- Visualizing the relationship between Flight Number and Orbit Type by Scatter Plot
- With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS. However, for GTO, it is difficult to distinguish between successful and unsuccessful landings as both outcomes are present.



Launch Success Yearly Trend

- Visualizing the launch success yearly trend by Line Plot
- Can see that the success rate since 2013 kept increasing till 2020



All Launch Site Names

- The names of the Unique Launch Sites in the Space Mission

```
[11]: %sql SELECT DISTINCT LAUNCH_SITE as "Launch_Sites" FROM SPACEXTBL;

 * sqlite:///my_data1.db
Done.
```

[11]:

Launch_Sites
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- ❑ The names of the Unique Launch Sites in the Space Mission
- ❑ Can see all the Landing Outcomes are all failure

```
%sql SELECT * FROM SPACEXTBL WHERE Launch_Site Like 'CCA%' LIMIT 5;
```

```
 * sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- The Total Payload Mass is 45596 kgs

```
%sql SELECT SUM(PAYLOAD_MASS__KG_) as "Total Payload Mass(Kgs)", Customer FROM 'SPACEXTBL' WHERE Customer = 'NASA (CRS)';
```

```
 * sqlite:///my_data1.db
Done.
```

Total Payload Mass(Kgs)	Customer
45596	NASA (CRS)

Average Payload Mass by F9 v1.1

- The average Payload Mass carried by Booster Version F9 v1.1 is 2534.6666 kgs

```
%sql SELECT AVG(PAYLOAD_MASS__KG_) as "Payload Mass Kgs", Customer, Booster_Version FROM 'SPACEXTBL' WHERE Booster_Version LIKE 'F9 v1.1%';
```

```
 * sqlite:///my_data1.db
Done.
```

Payload Mass Kgs	Customer	Booster_Version
2534.6666666666665	MDA	F9 v1.1 B1003

First Successful Ground Landing Date

- The date of the first successful Landing Outcome on Ground Pad was achieved on 2015-12-22

```
%sql SELECT MIN(DATE) FROM 'SPACEXTBL' WHERE "Landing_Outcome" = "Success (ground pad)";
```

```
 * sqlite:///my_data1.db
Done.
```

MIN(DATE)
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- The names of the Boosters which have success in Drone Ship

```
%sql SELECT DISTINCT Booster_Version, Payload FROM SPACEXTBL WHERE "Landing_Outcome" = "Success (drone ship)" AND PAYLOAD_MASS__KG_ > 4000 AND
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version	Payload
F9 FT B1022	JCSAT-14
F9 FT B1026	JCSAT-16
F9 FT B1021.2	SES-10
F9 FT B1031.2	SES-11 / EchoStar 105

Total Number of Successful and Failure Mission Outcomes

- There are 1 Failure in Flight, 99 successes and 1 success with unclear Payload status

```
%sql SELECT "Mission_Outcome", COUNT("Mission_Outcome") as Total FROM SPACEXTBL GROUP BY "Mission_Outcome";
```

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- Different Boosters carrying different maximum Payload Mass

```
%sql SELECT "Booster_Version", Payload, "PAYLOAD_MASS__KG_" FROM SPACEXTBL WHERE "PAYLOAD_MASS__KG_" = (SELECT MAX("PAYLOAD_MASS__KG_") FROM SPACEXTBL);
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version	Payload	PAYLOAD_MASS__KG_
F9 B5 B1048.4	Starlink 1 v1.0, SpaceX CRS-19	15600
F9 B5 B1049.4	Starlink 2 v1.0, Crew Dragon in-flight abort test	15600
F9 B5 B1051.3	Starlink 3 v1.0, Starlink 4 v1.0	15600
F9 B5 B1056.4	Starlink 4 v1.0, SpaceX CRS-20	15600
F9 B5 B1048.5	Starlink 5 v1.0, Starlink 6 v1.0	15600
F9 B5 B1051.4	Starlink 6 v1.0, Crew Dragon Demo-2	15600
F9 B5 B1049.5	Starlink 7 v1.0, Starlink 8 v1.0	15600
F9 B5 B1060.2	Starlink 11 v1.0, Starlink 12 v1.0	15600
F9 B5 B1058.3	Starlink 12 v1.0, Starlink 13 v1.0	15600
F9 B5 B1051.6	Starlink 13 v1.0, Starlink 14 v1.0	15600
F9 B5 B1060.3	Starlink 14 v1.0, GPS III-04	15600
F9 B5 B1049.7	Starlink 15 v1.0, SpaceX CRS-21	15600

2015 Launch Records

- List of the failed records which displays the month names, failure Landing Outcomes in Drone Ship, Booster Versions, Launch Site for the months in year 2015.

```
%sql SELECT SUBSTR(Date,6,2) AS Month, Booster_Version, Launch_Site FROM SPACEXTBL WHERE SUBSTR(Date,1,4) = '2015' AND LOWER(Landing_Outcome) LIKE '%failure%';
```

```
 * sqlite:///my_data1.db
Done.
```

Month	Booster_Version	Launch_Site
01	F9 v1.1 B1012	CCAFS LC-40
04	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- The count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20

```
%sql SELECT * FROM SPACEXTBL WHERE "Landing_Outcome" LIKE 'Success%' AND Date BETWEEN '2010-06-04' AND '2017-03-20' ORDER BY Date DESC;
```

```
 * sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2017-02-19	14:39:00	F9 FT B1031.1	KSC LC-39A	SpaceX CRS-10	2490	LEO (ISS)	NASA (CRS)	Success	Success (ground pad)
2017-01-14	17:54:00	F9 FT B1029.1	VAFB SLC-4E	Iridium NEXT 1	9600	Polar LEO	Iridium Communications	Success	Success (drone ship)
2016-08-14	5:26:00	F9 FT B1026	CCAFS LC-40	JCSAT-16	4600	GTO	SKY Perfect JSAT Group	Success	Success (drone ship)
2016-07-18	4:45:00	F9 FT B1025.1	CCAFS LC-40	SpaceX CRS-9	2257	LEO (ISS)	NASA (CRS)	Success	Success (ground pad)
2016-05-27	21:39:00	F9 FT B1023.1	CCAFS LC-40	Thaicom 8	3100	GTO	Thaicom	Success	Success (drone ship)

Section 3

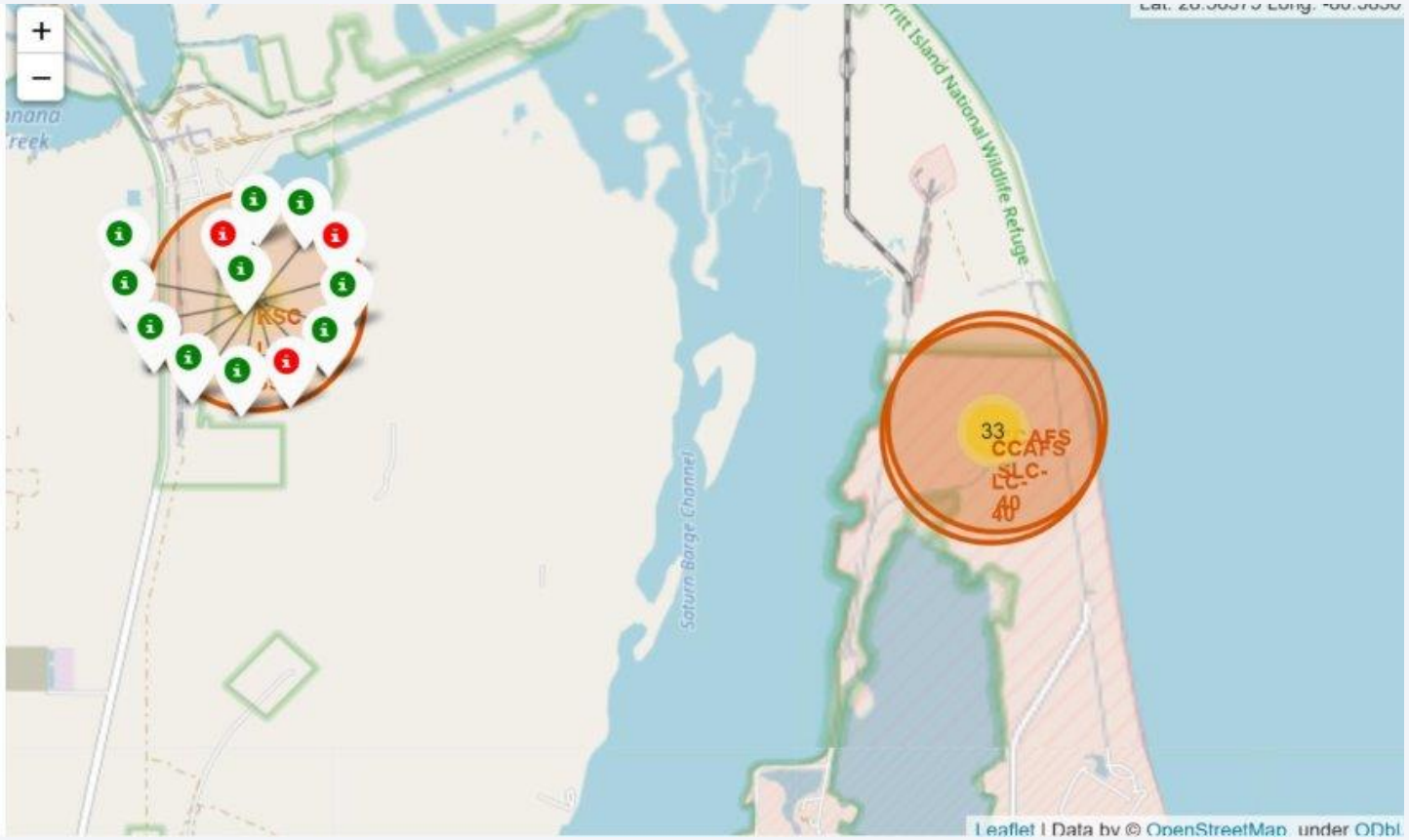
Launch Sites Proximities Analysis

All Launch Sites on a Global Map

- All launch sites in proximity to the Equator line
- All launch sites in remarkably proximity to the coast

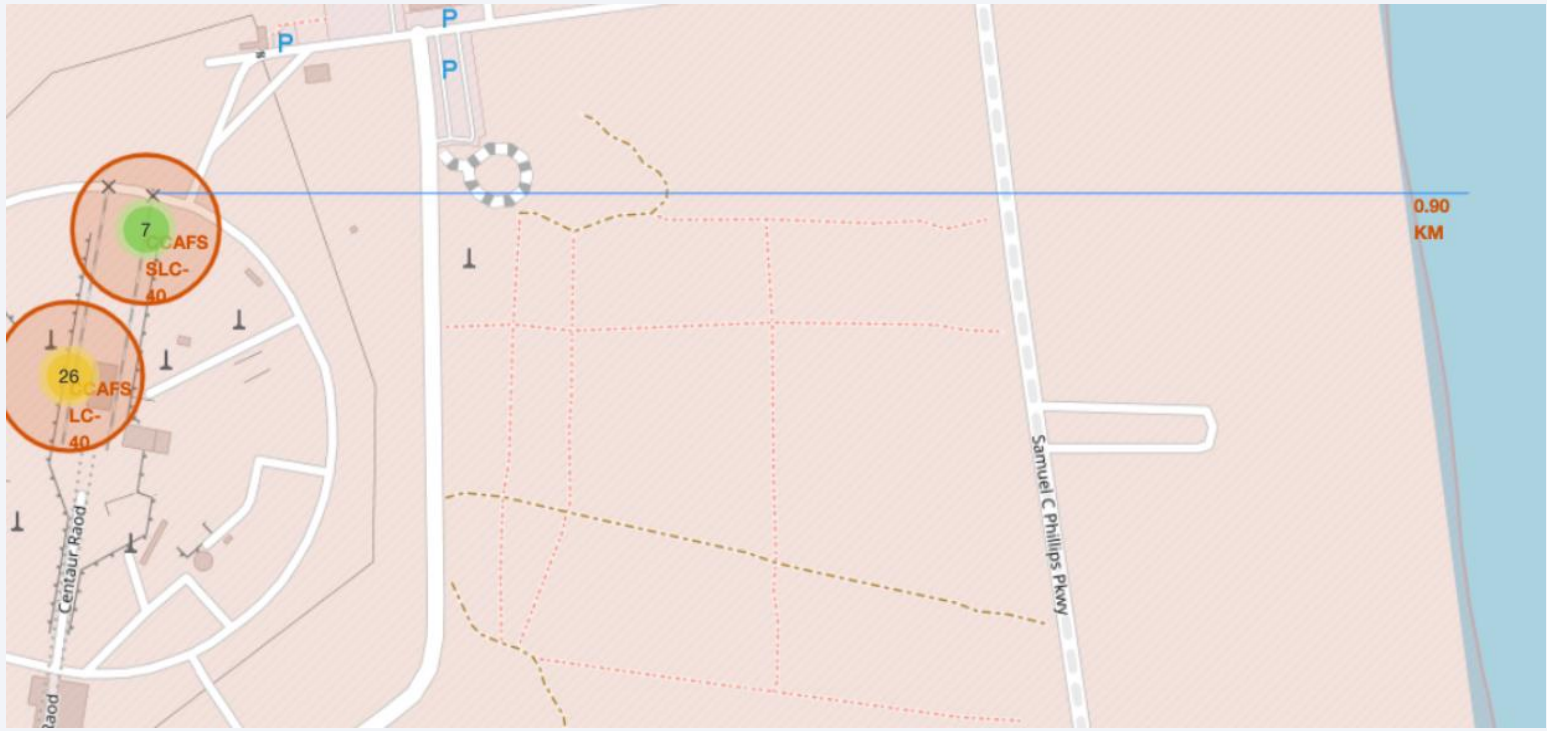
Launch Outcome Of Different Sites

- Left coast site has 10 Trails and right coast site has 46 Trails



The Proximity of All Launch Sites

- KSC LC-39A is 3.37 km far from the coast, and 5.11 km from the city
- CCAF SLC-40 is 1.91 km from the highway and 1.34km from the railway

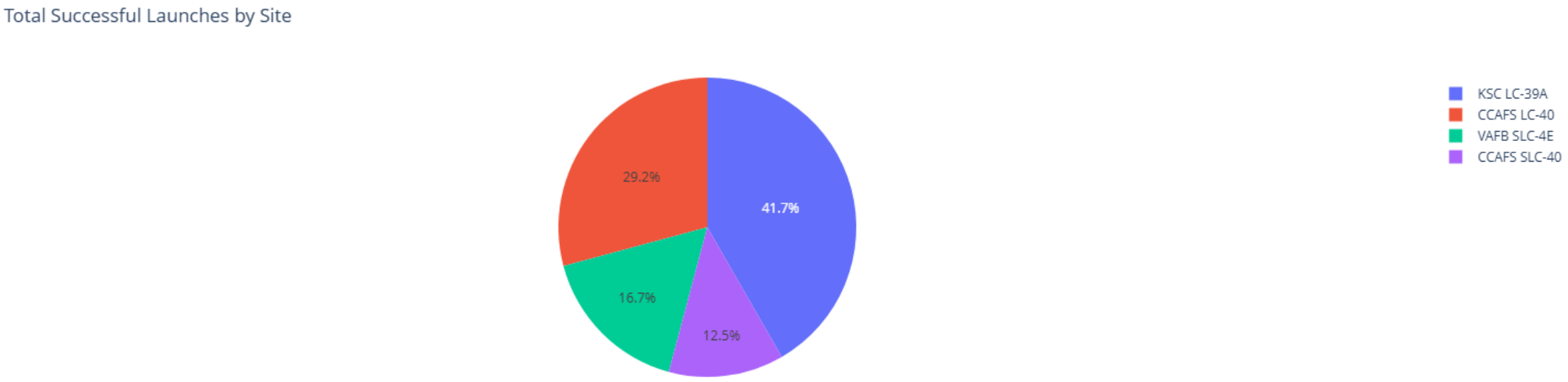


Section 4

Build a Dashboard with Plotly Dash

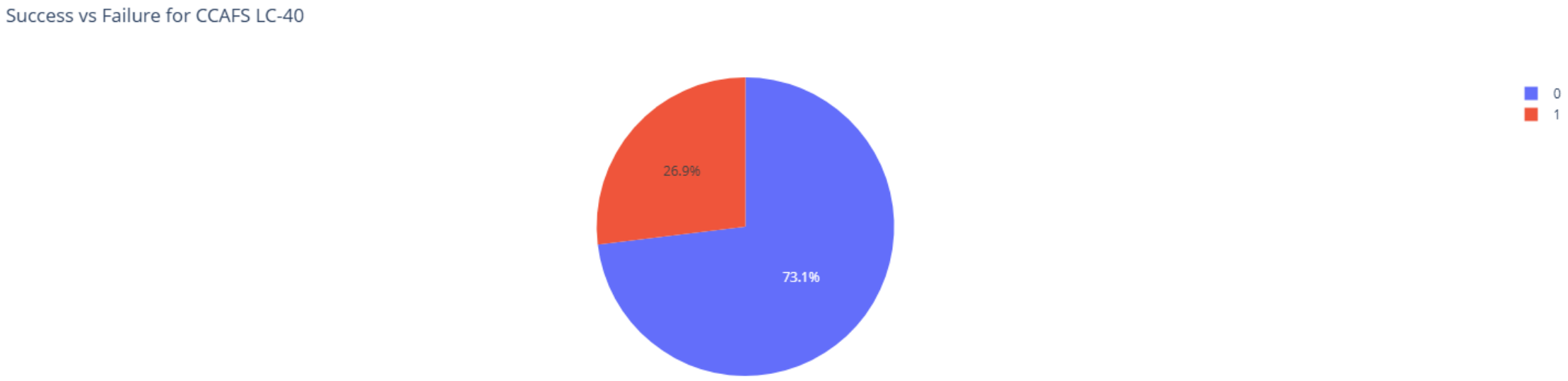
Total Successful launches by All Sites

- KSCLC39A:41.7%
- CCAFS LC-40: 29.2%
- VAFBSLC-4E: 16.7%
- CCAFS SLC-40: 12.5%



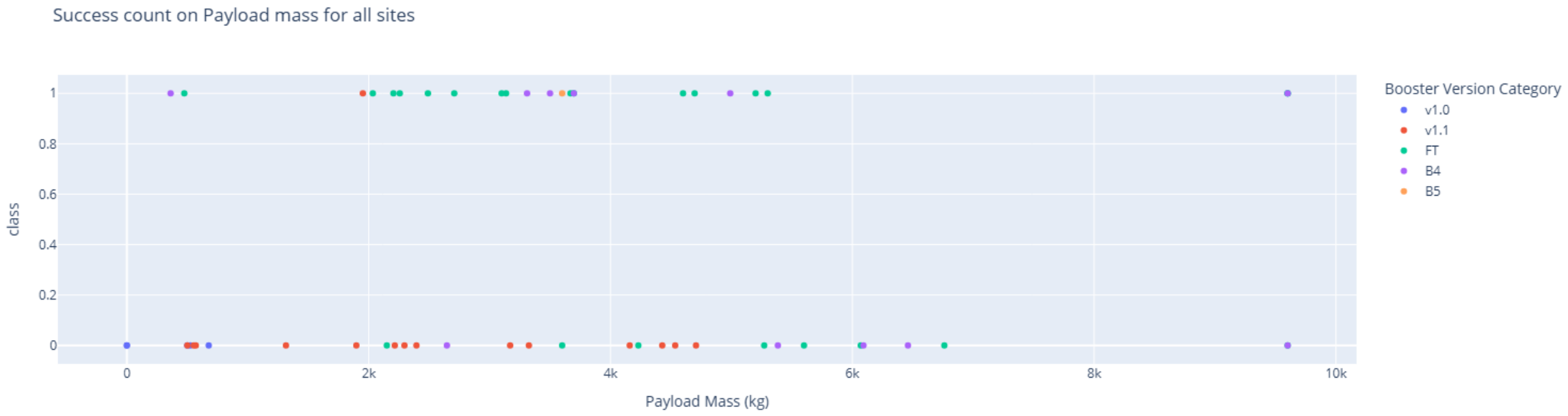
Highest Success Launch Ratio

- Success launch ratio is 73.1%



Payload vs Launch Outcome For All Sites

- For Launch site CCAFS LC-40 the booster version FT has the largest success rate from a payload mass of >2000kg



Section 5

Predictive Analysis (Classification)

Classification Accuracy

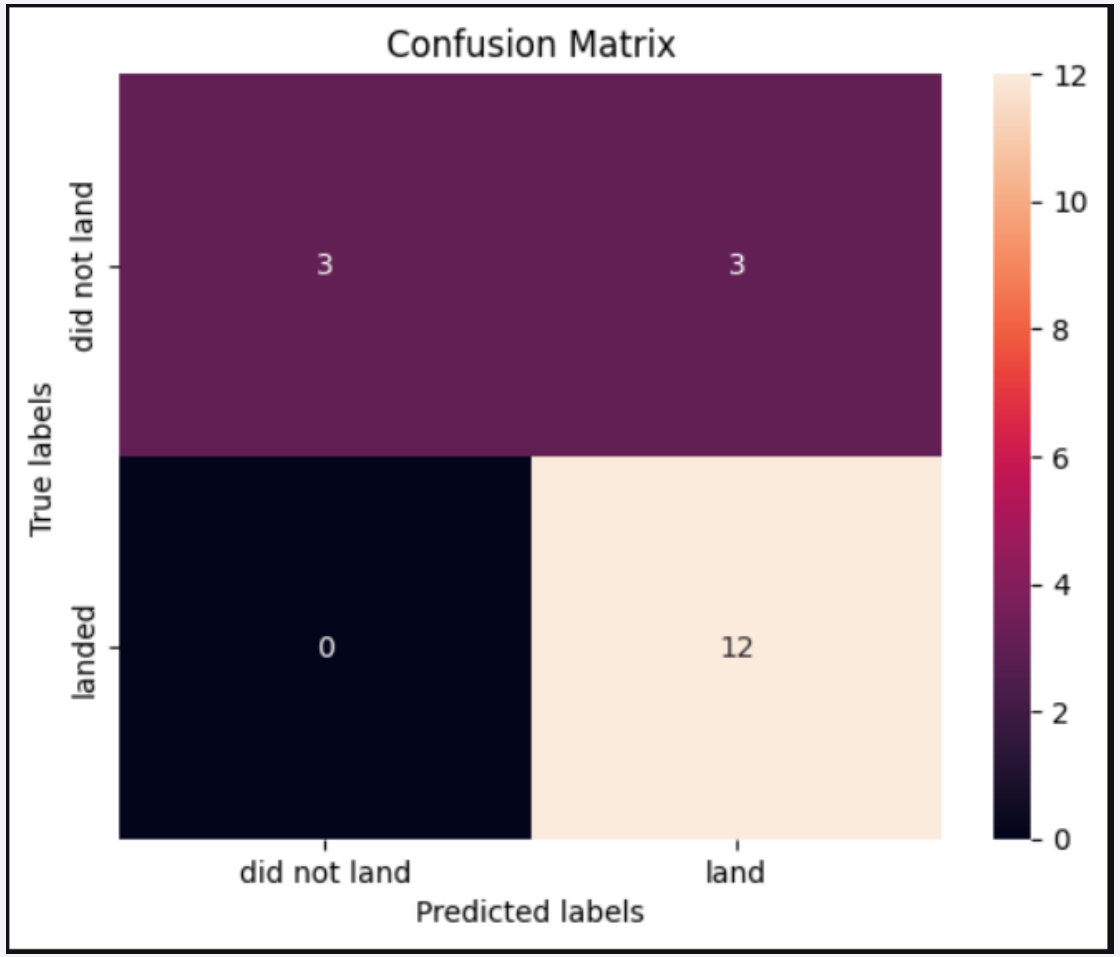
- We can see clearly that all the methods did equally well except the decision tree classifier

[67]:

	0
Method	Test Data Accuracy
Logistic_Reg	0.833333
SVM	0.833333
Decision Tree	0.722222
KNN	0.833333

Confusion Matrix

- All the 4 Classification model had the same confusion matrixes and were able equally distinguish between the different classes. The major problem is false positives for all the models.



Conclusions

- There is a correlation between launch site and success rate Payload mass is also associated with the success rate.: the more massive the payload, the less likely the first stage will return
- For orbit type, SO has the least success rate while ES-L1, GEO, HEO and SSO have the highest success rate According to the yearly trend
- There has been an increase in the success rate since 2013 kept increasing till 2020
- Different launch sites have different success rates. CCAFS LC-40, has a success rate of 60 %, while KSC LC-39A and VAFB SLC 4E has a success rate of 77%.

Thank you!